INFOSYS SPRINGBOARD

DATACO
SUPPLY CHAIN ANALYTICS

An Interactive Data Analytics and Business Intelligence Approach to Supply Chain Performance

FINAL PROJECT REPORT

Infosys Springboard Virtual Internship

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Project Theme
From Data Cleaning to Actionable Supply Chain Insights

EXECUTIVE SUMMARY

Abstract

This project presents an interactive analytics solution for the DataCo supply chain dataset. The work follows a complete data-to-decision workflow: dataset understanding, data preprocessing, descriptive analysis, KPI development, visualization and Power BI dashboard creation. The documented raw project state contains 2,599 records and 24 columns, including 138 rows with missing values before preprocessing. The data covers sales, orders, products, inventory, delivery outcomes, shipping modes, customer segments and geographic dimensions.

Python, Pandas and NumPy were used for inspection, missing-value treatment, duplicate checking, date conversion and standardization. The final analysis examines sales trends, delivery status, regional performance, product and inventory performance, categories, cities and shipping efficiency. The Power BI solution contains four dashboards: Executive Overview, Warehouse Efficiency, Inventory & Product Performance, and Delivery & Regional Performance.

Key project measures include approximately 6.13M total sales, 3K orders, 220.13K profit, 3.95 average shipping/delivery days, 32.13% on-time delivery, approximately 40K inventory units and 11.65M inventory value. The analysis highlights delivery reliability as a major improvement area and shows regional differences, with South as the strongest displayed sales region. The project is descriptive and dashboard-oriented; predictive modelling and transportation-cost analysis are not claimed as completed.

Keywords

DataCo; Supply Chain Analytics; Data Preprocessing; Python; Pandas; NumPy; Power BI; KPI; Inventory; Delivery Performance; Regional Analysis; Business Intelligence.

REPORT STRUCTURE

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Reading the report

The chapters are arranged as one continuous project story—from raw data quality to analytical insight and management action. No certificate, declaration or acknowledgement pages are included.

FIGURES & TABLES

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CHAPTER 1 | INTRODUCTION

1.1 Introduction

Supply-chain operations generate large volumes of interconnected records about orders, customers, products, inventory, shipping and delivery. Analytics provides a way to organize these records into evidence that can support planning, monitoring and operational decisions. The DataCo project applies this approach to order-level supply-chain information and develops a visual business-intelligence solution.

1.2 Supply Chain Management

Supply chain management requires coordination of products, information and operational activities across the flow from order placement to customer delivery. From an analytics perspective, this creates measurable dimensions such as order volume, sales, profit, delivery status, shipping duration, inventory position, customer segment and geography.

1.3 Role of Analytics

- Sales analytics provides visibility into trends, markets, regions and categories.
- Delivery analytics highlights late, on-time and advance-shipping patterns.
- Inventory analytics provides product-level visibility of stock position.
- Geographic analysis reveals differences across markets, countries, regions and cities.
- Shipping-mode analysis provides comparative evidence about delivery performance.

1.4 Project Background

The project combines milestone work on data preprocessing, analysis and interactive dashboards into one integrated solution. The central principle is that reliable visualization depends on a clean and consistently structured analytical dataset.

Project focus

The completed work is primarily descriptive analytics and dashboard-based decision support. It does not claim predictive analytics or machine learning unless explicitly supported by the project materials.

CHAPTER 2 | PROBLEM STATEMENT, OBJECTIVES & SCOPE

2.1 Problem Statement

The challenge is to convert operational supply-chain records into a reliable, understandable and interactive performance-monitoring solution. Raw records may contain missing values, inconsistent formats or duplicate observations, while a large table by itself does not provide an efficient management view. The project addresses this by combining preprocessing, KPI analysis and Power BI visualization.

2.2 Objectives

- Understand the dataset structure, columns and data types.
- Identify and treat missing values using documented methods.
- Check duplicate records and prepare a consistent analytical dataset.
- Standardize column names and category labels and convert date fields.
- Analyze sales, orders, delivery, inventory, products, categories, regions and cities.
- Compare shipping performance using actual and scheduled shipping days, delivery status and risk.
- Develop interactive Power BI dashboards for management and operational decision support.

2.3 Scope

The scope covers order-level sales, profit, quantity, product, inventory, customer, market, region, country, shipping and delivery information. The analysis includes descriptive KPIs and visual comparisons. Transportation expenditure is outside the scope because the dataset does not provide a direct transportation-cost field.

2.4 Business Questions

- How does sales performance vary over time and across regions or markets?
- Which delivery-status patterns require operational attention?
- How do products, categories and inventory contribute to performance?
- How do shipping modes and regions differ in delivery outcomes?
- Which combinations of region, customer segment and shipping mode show higher delivery risk?

CHAPTER 3 | DATASET DESCRIPTION & DATA UNDERSTANDING

3.1 Dataset Overview

The project documentation identifies the DataCo Supply Chain Raw Dataset as the main analytical source. It contains 2,599 rows and 24 columns, with 138 rows containing missing values before preprocessing. The supplied cleaned workbook represents the prepared state used for analysis.

Dataset characteristic	Verified project information
Rows	2,599
Columns	24
Missing-value state before preprocessing	138 rows with missing values
Business grain	Order-level supply-chain records
Markets	LATAM, USCA, Pacific Asia, Africa, Europe
Core domains	Sales, orders, shipping, delivery, products, inventory, customers and geography

3.2 Geographic Coverage

The project uses Market, Order Country, Order Region and Customer City to support analysis at several geographic levels. The five documented markets are LATAM, USCA, Pacific Asia, Africa and Europe.

3.3 Business Relevance

The combination of financial, operational, product and geographic attributes allows the same dataset to support management KPIs and operational drill-downs. This makes it suitable for a dashboard-based supply-chain performance study.

Figure 3.1: DataCo Dashboard / Project Overview

3.4 Major Dataset Attributes

Attribute	Meaning	Business use
Order ID	Order identifier	Record identification
Type	Order/payment type	Order-type comparison
Days for shipping (real)	Actual shipping duration	Delivery speed
Days for shipment (scheduled)	Scheduled duration	Plan-versus-actual comparison
Delivery Status	Advance/on-time/late status	Delivery monitoring
Late Delivery Risk	Risk indicator	Risk monitoring

Category Name	Product category	Category analysis
Customer City	Customer location	City analysis
Customer Segment	Customer segment	Segment comparison
Department Name	Department	Drill-down
Market	Global market	Market comparison
Order Country	Country	Geographic drill-down
Order Date	Order date	Time-series analysis
Order Item Quantity	Quantity ordered	Volume analysis
Sales	Sales value	Revenue analysis
Unit Price	Unit price	Price context
Order Item Total	Order item value	Order-value context
Order Profit Per Order	Profit per order	Profitability
Order Region	Region	Regional comparison
Order Status	Order status	Order monitoring
Product Name	Product	Product analysis
Shipping Date	Shipping date	Shipping timeline
Shipping Mode	Shipping method	Mode comparison
Inventory Stock (Units)	Inventory quantity	Stock monitoring

The attribute list above reflects the 24-field structure of the supplied DataCo workbook. Date fields are treated as dates after preprocessing.

CHAPTER 4 | DATA PREPROCESSING

4.1 Preprocessing Workflow

The preprocessing stage converts raw records into a consistent analytical dataset. The documented tools are Python, Pandas and NumPy.

1. Load	2. Inspect	3. Clean	4. Standardize	5. Prepare
Excel dataset	Shape, types, missing values	Missing values & duplicates	Dates, labels, column names	Analysis-ready data

4.2 Loading and Initial Inspection

The dataset was loaded from Excel and inspected for shape, column structure and data types. The documented initial size was 2,599 records by 24 columns.

4.3 Missing-Value Treatment

The raw project state included 138 rows with missing values. Numerical missing values were treated using median imputation where documented. Categorical missing values were treated using mode imputation where documented.

4.4 Duplicate Checking

Duplicate records were checked and removed where applicable. The supplied cleaned workbook contains 2,599 records and zero exact duplicate rows.

4.5 Date Conversion

Order Date and Shipping Date were converted into proper datetime format so that time-based analysis and shipping timelines could be handled consistently.

4.6 Standardization

Column names and category labels were standardized as documented in the preprocessing milestone. Consistency at this stage reduces grouping errors during visualization.

4.7 Representative Python Code

```
import pandas as pd
import numpy as np
df = pd.read_excel('dataco.xlsx')
print(df.shape)
print(df.dtypes)
df['Sales'].fillna(
    df['Sales'].median(),
    inplace=True)

df['Category Name'].fillna(
    df['Category Name'].mode()[0])
df.drop_duplicates(inplace=True)
df['Order Date'] = pd.to_datetime(df['Order Date'])
df['Shipping Date'] = pd.to_datetime(df['Shipping Date'])
```

4.8 Preprocessing Summary

Problem	Method	Purpose	Outcome
Missing numerical values	Median imputation	Preserve central tendency	Handled
Missing categorical values	Mode imputation	Use most frequent class	Handled
Duplicate records	Check and remove	Avoid repeated observations	None in cleaned workbook
Date fields	Datetime conversion	Enable time analysis	Prepared
Inconsistent labels	Standardization	Consistent grouping	Prepared

Quality checkpoint
The supplied cleaned workbook is populated across the 24 fields and contains no exact duplicate rows. The documented 138 missing-value rows refer to the preprocessed project state.

CHAPTER 5 | DATA ANALYSIS & VISUALIZATION

5.1 Analysis Methodology

After preprocessing, the project transitions from data quality work to descriptive analysis. The analysis uses KPIs and visual comparisons to understand sales, orders, delivery, inventory, products, categories and geography.

5.2 KPI Analysis

KPI	Project value	Business interpretation
Total Sales	6.13M	Overall commercial scale
Total Orders	3K	Overall order volume
Total Profit	220.13K	Profitability context
Average Shipping/Delivery Days	3.95	Delivery-duration benchmark
On-Time Delivery	32.13%	Major service-performance improvement area

Late Delivery Rate	~41.7–42%	Substantial late-delivery share
Total Inventory	~40K units	Stock-monitoring scale
Inventory Value	11.65M	Commercial importance of inventory

5.3 Sales Trend Analysis

The sales trend provides time-based visibility into stronger and weaker periods. Rather than relying on a single sales total, the visual supports identification of changes over time and helps management investigate periods requiring further explanation.

Figure 5.1: Sales Trend Analysis

5.4 Delivery Status Analysis

The cleaned workbook contains 1,083 late deliveries, 835 shipping-on-time records and 681 advance-shipping records. These correspond to approximately 41.7%, 32.1% and 26.2% of the 2,599 records respectively.

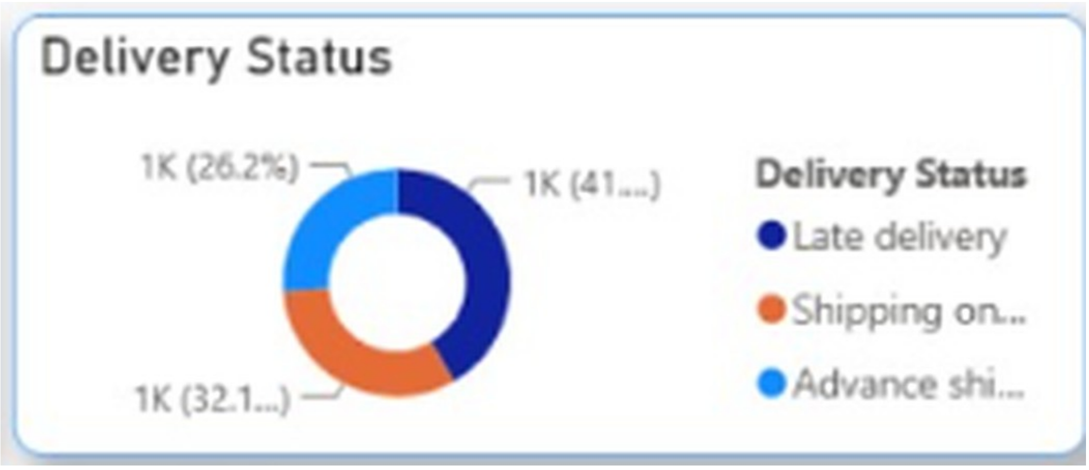

Figure 5.2: Delivery Status Analysis

5.5 Regional Sales Performance

Regional sales comparisons show South as the strongest displayed sales region, followed by East, West, North and Central. The value of the comparison is that it identifies where commercial performance differs geographically.

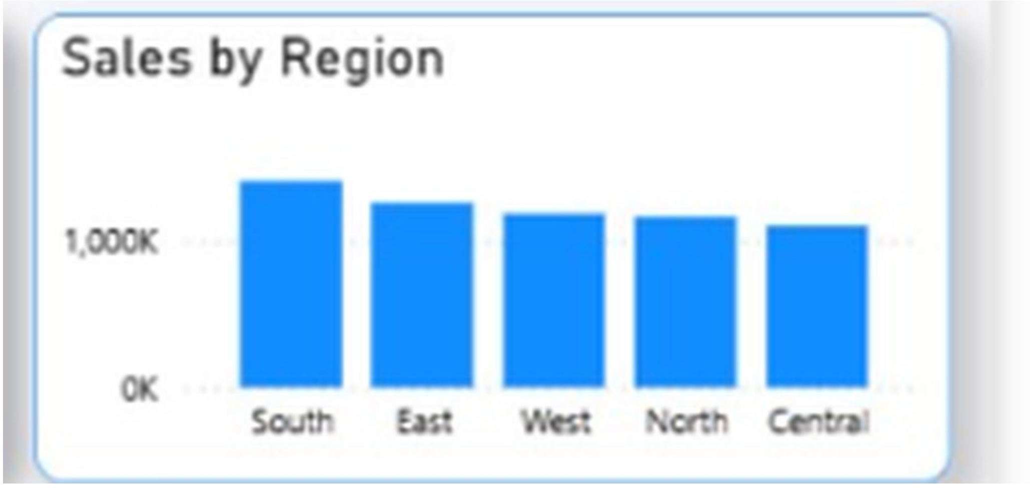

Figure 5.3: Regional Sales Performance

5.6 Regional Delivery Performance

Regional delivery analysis compares Advance shipping, Late delivery and Shipping on time across South, East, North, Central and West, allowing users to identify regional delivery patterns that merit investigation.

CHAPTER 6 | POWER BI DASHBOARD DEVELOPMENT

6.1 Dashboard Architecture

The Power BI solution follows the sequence: cleaned dataset → measures and KPIs → visual comparisons → filters and drill-downs → business interpretation. The dashboards are designed to move from management-level summaries to operational detail.

6.2 Dashboard 1 – Executive Overview

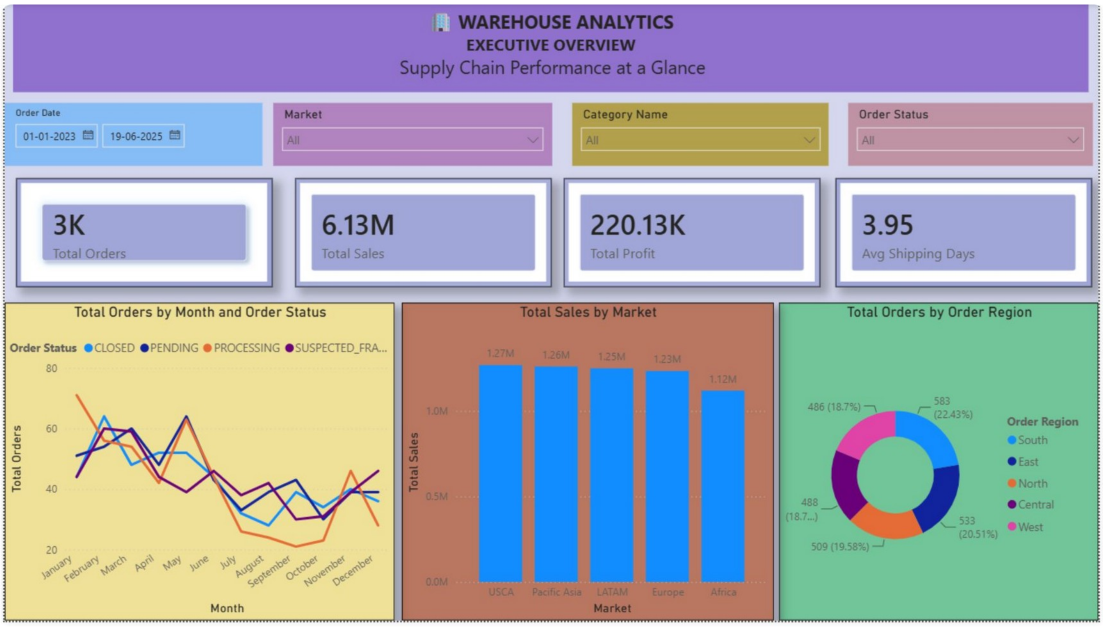

Figure 6.1: Executive Overview Dashboard

The Executive Overview combines Total Orders (3K), Total Sales (6.13M), Total Profit (220.13K) and Average Shipping Days (3.95). It also provides sales-by-market and orders-by-region views, giving management a compact starting point for performance review.

Management value

The dashboard supports rapid identification of overall performance and provides the first level of drill-down into market and regional differences.

6.3 Dashboard 2 – Warehouse Efficiency

Figure 6.2: Warehouse Efficiency Dashboard

Warehouse Efficiency focuses on orders, shipping efficiency and delivery performance. Its headline measures include 3K total orders, 3.95 average shipping days and approximately 42% late-delivery rate. Monthly orders, monthly shipping days, delivery status, market and shipping mode provide operational context.

6.4 Interactive Analysis

- Monthly views provide temporal context for operational performance.
- Delivery-status views distinguish advance, on-time and late outcomes.
- Market and shipping-mode comparisons help narrow the location or method requiring attention.
- Interactive filters allow users to investigate selected subsets rather than only overall totals.

6.5 Dashboard 3 – Inventory & Product Performance

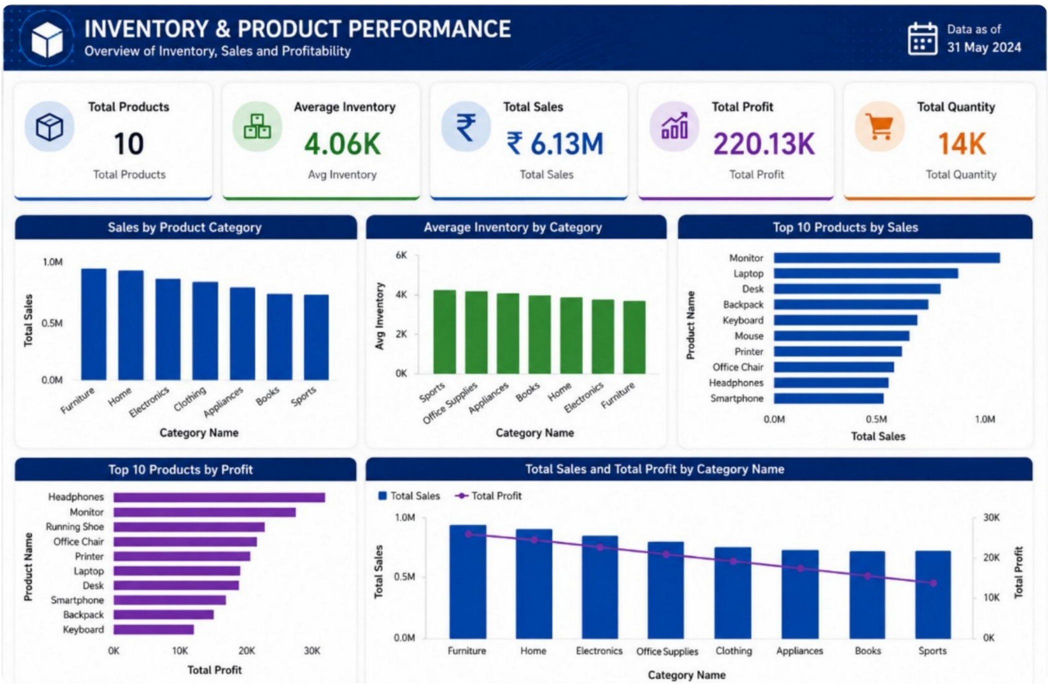

Figure 6.3: Inventory & Product Performance Dashboard

The dashboard combines product, inventory, sales and profitability. Its displayed KPIs include 10 total products, 4.06K average inventory, 6.13M total sales and 14K total quantity. Product and category views help connect stock position with commercial performance.

6.6 Decision Support

Inventory should not be monitored in isolation. Product sales, profit and category performance provide context for deciding which products deserve closer stock monitoring and which categories contribute more strongly to commercial results.

6.7 Dashboard 4 – Delivery & Regional Performance

Figure 6.4: Delivery & Regional Performance Dashboard

This dashboard focuses on delivery reliability. It displays 3K total orders, approximately 41.7% late delivery, approximately 32.1% on-time delivery and an average delivery gap of 0.45. The supporting visuals compare markets, regions, countries, shipping days and shipping modes.

6.8 Dashboard Comparison

Dashboard	Primary purpose	Key dimensions
Executive Overview	Management summary	Sales, orders, profit, market, region
Warehouse Efficiency	Operational monitoring	Orders, shipping days, status, market, mode
Inventory & Product	Stock and commercial decisions	Products, inventory, sales, category, profit
Delivery & Regional	Reliability and geographic risk	Late rate, on-time %, region, country, mode

CHAPTER 7 | ADVANCED BUSINESS ANALYSIS

7.1 Regional Performance Scoreboard

Figure 7.1: Regional Performance Scoreboard

The regional scoreboard combines sales, orders, inventory and delivery performance. South is identified as the strongest displayed region by sales, while the broader comparison shows variation among regions. Filters such as Order Country and Order Status enable deeper investigation.

7.2 Shipping Efficiency

Because the dataset has no direct transportation-cost field, shipping efficiency is assessed through Shipping Mode, actual shipping days, scheduled shipping days, Delivery Status, Late Delivery Risk, Order Region and Order Country.

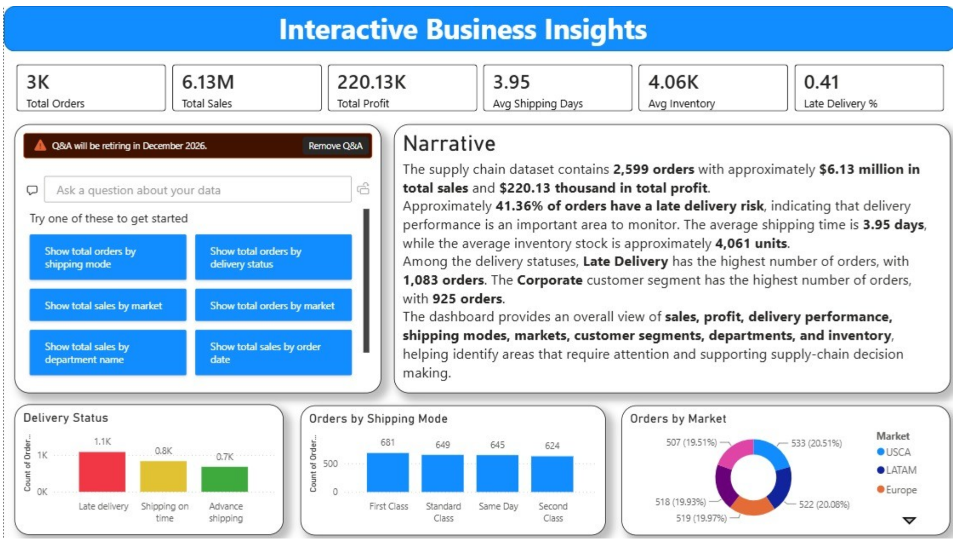

Figure 7.2: Shipping Efficiency Analysis

Interpretation boundary
This analysis provides logistics visibility, not actual transportation expenditure or cost-efficiency measurement.

7.3 Route and Logistics Drill-Down

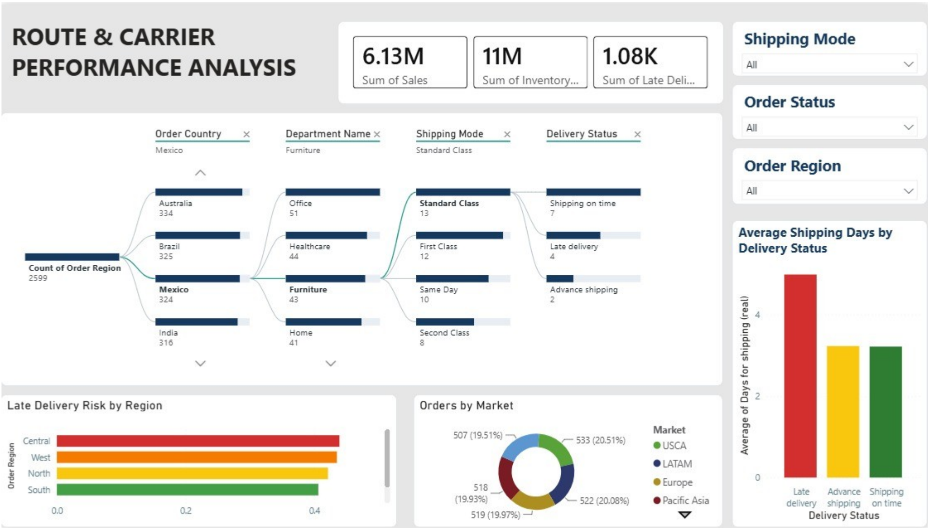

Figure 7.3: Route & Logistics Drill-Down

The project uses the drill-down sequence Order Country → Department → Shipping Mode → Delivery Status. This allows a broad geographic issue to be narrowed into a more specific operational combination. Carrier-level performance is not claimed because the source materials do not establish a carrier-performance field.

7.4 Delivery Risk Heat Map

Figure 7.4: Delivery Risk Heat Map

The heat map compares Order Region × Customer Segment × Shipping Mode. It supports identification of combinations associated with higher delivery time or risk. The project reports approximately 1.08K late-delivery-risk records.

7.5 Customer Segments and Markets

Customer Segment includes Consumer, Corporate and Home Office. Market includes LATAM, USCA, Pacific Asia, Africa and Europe. These dimensions provide structured comparison without implying unsupported causality.

CHAPTER 8 | KEY FINDINGS & BUSINESS INSIGHTS

8.1 Key Findings

Finding	Evidence	Business meaning
Sales performance	6.13M total sales	Overall commercial scale
Order activity	3K total orders	Management view of order volume
Profitability	220.13K total profit	Adds profit context to sales
Delivery speed	3.95 average days	Benchmark for delivery duration
On-time delivery	32.13%	Major service-performance gap
Late delivery	~41.7–42%	Substantial late-delivery share
Inventory	~40K units	Large stock position requiring monitoring
Inventory value	11.65M	Material commercial exposure in inventory
Regional performance	South strongest displayed sales region	Geographic variation requires comparison

8.2 Integrated Business Meaning

The findings become more useful when interpreted together. Sales and profit indicate commercial performance; inventory indicates stock position; shipping days and delivery status indicate service execution; and region, market, product and shipping mode provide the dimensions needed to investigate differences. The dashboards therefore create a connected decision-support view rather than a collection of isolated charts.

Figure 7.5: Inventory & Delivery Performance

CHAPTER 9 | RECOMMENDATIONS

9.1 Practical Recommendations

Priority	Evidence / problem	Recommended action	Expected benefit
Improve delivery reliability	Low on-time delivery and high late delivery	Monitor delivery status and actual-versus-scheduled days; investigate recurring late patterns.	Better prioritization of delivery issues.
Monitor late-delivery risk	~1.08K late-risk records	Use region × segment × shipping-mode risk views for focused monitoring.	Earlier identification of high-risk combinations.
Compare shipping modes	Delivery outcomes vary by mode	Compare actual/scheduled days and delivery status by mode.	Evidence-based mode decisions.
Strengthen regional monitoring	Regional performance varies	Review sales and delivery KPIs together by region/country.	More targeted regional management.
Strengthen inventory monitoring	~40K units; 11.65M value	Review inventory alongside product sales and category performance.	Better stock visibility.
Monitor product performance	Product-level sales/profit/inventory views available	Prioritize products needing closer review.	Focused product decisions.
Use dashboards routinely	Four complementary interactive views	Use KPI cards for monitoring and drill-downs for investigation.	Faster, consistent review.
Improve route/market monitoring	Country → Department → Mode → Status hierarchy	Investigate specific operational combinations.	More precise problem localization.

Important
No financial savings or transportation-cost reductions are claimed because those quantities are not supported by the dataset.

CHAPTER 10 | LIMITATIONS & FUTURE SCOPE

10.1 Limitations

- No direct transportation-cost field is available; shipping efficiency is therefore assessed using operational shipping and delivery metrics.
- The project is primarily focused on data preprocessing, descriptive analysis, visualization and dashboard-based decision support.
- The uploaded materials do not establish a completed predictive or machine-learning model.
- Dashboard interpretations depend on selected filters and dimensions and should not be treated as causal explanations.

10.2 Data Limitations

The documented raw project state contained missing values, while the supplied cleaned workbook is populated across its 24 columns and has no exact duplicate rows. Category naming contains capitalization variants in the workbook, so equivalent categories should be treated consistently during reproduction.

10.3 Future Scope

- Predictive delivery-risk modelling
- Demand and inventory forecasting
- Anomaly detection
- Logistics and route optimization
- Real-time dashboard integration
- Transportation-cost data integration
- Carrier-level performance data integration

Future scope
These are proposed extensions only; they are not presented as completed capabilities of the current project.

CHAPTER 11 | CONCLUSION & REFERENCES

11.1 Conclusion

The DataCo Supply Chain Analytics project demonstrates a complete transformation from operational supply-chain records to an interactive decision-support solution. It begins with dataset understanding and data-quality assessment, followed by documented preprocessing using Python, Pandas and NumPy. Missing values, duplicate records, dates and inconsistent labels are addressed before analysis.

The cleaned data is then used for KPI development and descriptive visualization across sales, orders, delivery, inventory, products, categories, regions and cities. Four Power BI dashboards organize these insights into management, warehouse, product/inventory and delivery/regional perspectives. The project reports 6.13M sales, 3K orders, 220.13K profit, 3.95 average delivery/shipping days, 32.13% on-time delivery, approximately 40K inventory units and 11.65M inventory value.

The strongest operational message is the need to improve delivery reliability. Regional comparisons, shipping-mode analysis and delivery-risk views provide the evidence needed to prioritize further investigation. Overall, the project fulfills its central theme: “From Data Cleaning to Actionable Supply Chain Insights.”

References

- Infosys Springboard Virtual Internship – DataCo Supply Chain Analytics project materials supplied for this report.
- DataCo Supply Chain Raw Dataset / cleaned DataCo workbook supplied for the project.
- Project milestone presentation covering data analysis, preprocessing and advanced business analysis.
- Project milestone presentation covering interactive Power BI dashboard development.
- Python, Pandas and NumPy – tools documented in the project preprocessing materials.
- Power BI – dashboard and interactive visualization platform documented in the project materials.

Final statement

The solution provides a descriptive, interactive framework for monitoring supply-chain performance and prioritizing operational improvement.